Importance of Machine Learning and Deep Learning Algorithms in Earthquake Prediction: A Review

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***Abstract*— Earthquakes are the leading natural disasters that have caused loss of life and property since the formation of the world. Machine learning and deep learning are frequently used in studies for earthquake prediction. This article consists of a compilation of studies using machine learning and deep learning algorithms. In the article, studies on topics such as earthquake magnitude estimation, signal discrimination, electron density estimations in the ionosphere, examination of radon gas anomalies using machine learning and deep learning algorithms are included. The studies in this paper show that Deep Learning algorithms are used more frequently in earthquake forecasting. It is expected that Deep Learning will provide more successful results in future studies due to its ability to work with larger data sets compared to Machine Learning and its ability to improve itself from errors.**

Keywords*— *earthquake, machine learning, deep learning, early earthquake systems,

I. Introduction

Many natural disasters such as floods, earthquakes, and fires occur every year around the world. Undoubtedly earthquakes cause the most damage to societies, people, and states in terms of economic and loss of life. Earthquakes with a magnitude of 7.7 and 7.6, which occurred on the same day in Turkey's Kahramanmaraş province on February 06, 2023, caused great destruction in Turkey and Syria and ended the lives of thousands of people. In the report published by the World Bank on February 20, 2023, shortly after the Kahramanmaraş earthquake, it is estimated that the damage caused by the earthquakes in Turkey is 34.2 billion USD [1]. Globally, it has been determined that the earthquakes that have occurred in the last two decades have caused economic damages upward of 650 billion USD [2].

In addition to the material damage caused by earthquakes, millions of people can become disabled due to earthquakes and have to continue their lives with a disability. With the damage of underground water pipes due to the damage caused by the earthquake, it can be difficult for people in the earthquake area to reach clean water in the first days, which can cause epidemics. The collapse of the energy infrastructure also complicates the search and rescue efforts in earthquake zones, greatly restricting the heating needs of the people, especially due to earthquakes that occur in the winter months. The effect of the earthquakes on roads, airports and railways also greatly impact the ease by which aid can reach the earthquake center. Such damages to the transportation infrastructure can prevent the delay of the proper equipment and personnel who will participate in the rescue efforts. It also limits the flow of necessities such as food, cleaning material, and shelter aids for the affected population. All these difficulties are not only limited to a few days after the earthquake, but their psychological effects continue for many years. Such natural disasters additionally have direct effect on the education system of a region causing lower school grades in the affected areas. [3].

In order to alleviate and prevent the various burdens caused by earthquakes on societies and states, studies are carried out on the prediction of earthquakes and the development of early warning systems. The integration of these studies with artificial intelligence algorithms enables more precise predictions to be made. In this article, examples of machine learning and deep learning algorithms in earthquake prediction studies are given.

II. Machine Learning, Deep Learning And Earthquakes

Machine learning (ML) is one of the sub-branches of artificial intelligence that allows us to make predictions and classifications with the help of algorithms without requiring explicit programming with the help of historical information or instantaneous information. Machine learning includes supervised learning, unsupervised learning, semi-supervised learning and reinforcement learning. In particular, supervised learning is used in classification and regression problems where labeled datasets are used, and unsupervised learning using unlabeled datasets is used in clustering and size reduction problems.

Deep Learning is a sub-branch of machine learning derived from Neural Networks, mostly used in image processing, speech recognition, natural language processing [4] time series problems, which can learn incrementally from the layers that make up the network or the entire network as a whole, can perform operations such as feature extraction without human intervention, and can be trained on a data set without starting from the beginning [5].

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In the studies, it has been observed that machine learning has made progress with supervised learning algorithms in detection, arrival time measurement, phase association, location and characterization tasks, which are the stages of monitoring earthquakes [6]. However, in some cases, the predictions or classifications made by supervised learning with labeled data are not suitable for revealing unknown patterns. In such cases, unsupervised learning techniques can be used to cluster seismic wave data. For example, identifying patterns in seismic wave data with unsupervised learning revealed precursor signals before the landslide and tsunami in Greenland in 2017 [6]. In the past, distinguishing seismic signals was difficult or impossible due to limited technology. Today, however, machine learning algorithms can distinguish local earthquake signals from various disturbance signals [7]. Machine learning techniques are used to distinguish acoustic signals. Studies are carried out on earthquakes, tsunami events, estimation of tectonic event magnitudes by distinguishing acoustic signals. [8].

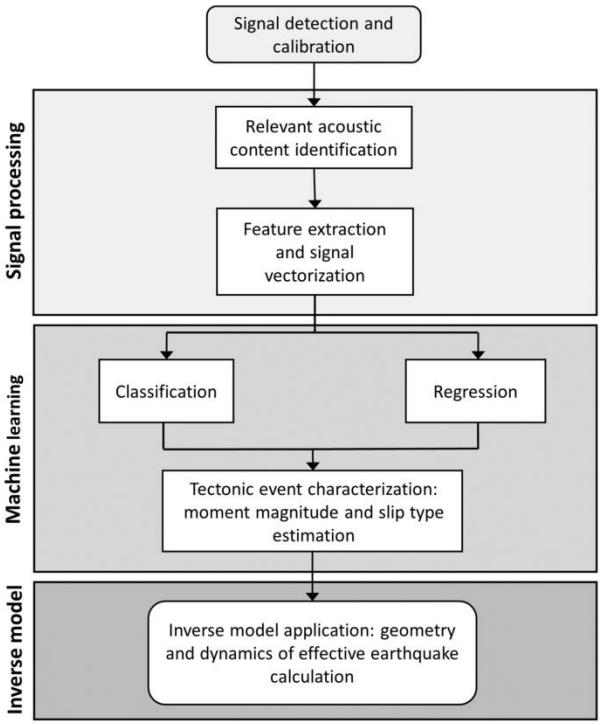

Fig. 1. Tectonic event characterization with ML Techniques [8].

Machine learning is used in almost every field. In earthquake prediction methods, many algorithms are used according to the problem.

III. Methods

In December 2015, the Earth Sciences and Geohazard Risk Reduction Division of UNESCO launched the International Platform for Earthquake Early Warning Systems (IP-EEWS), which promotes international cooperation and scientific information exchange. Ten countries are available on this platform, including China, Germany, Italy, Japan, Mexico, Romania, Spain, Switzerland, Turkey, United States [9]. The development of early warning systems (EEW) is important in preventing loss of life and property, preventing damage to infrastructures such as oil, natural gas, water pipelines, communication infrastructure, preventing enter to buildings, tunnels, bridges, plays a role. EEW includes the detection and location of the earthquake event, estimating the magnitude of the earthquake, estimating the tremors at various distances, and collecting and using this data whether to give an alarm in the warning system [10]. However, these systems do not predict the exact time of earthquakes in advance and provide warning by quickly detecting earthquakes [11]. There is no precise method for predicting earthquakes that gives the time information of the earthquake. Among the reasons for this are the fact that all the factors that will create the earthquake have not been discovered and the seismic features have complex and non-linear structures [12].

The basic algorithms used in the studies in this article are briefly described below and some prediction methods using machine learning and deep learning in earthquake predictions are given.

A. Machine Learning and Deep Learning Algorithms

Before discussing the algorithms below, it is necessary to define the terms such as classification, regression, time series and clustering.

- Classification: It is the process of assigning items with similar characteristics to a group. In classification, the dependent (target) variable to be predicted must be qualitative. Sometimes the quantitative variables given for the dependent variable refer to a class, not a numerical quantity. Example: 0: negative, 1: positive.
- Regression: Shows the strength of the relationships between a dependent and one or more independent variables. Quantitative expressions are used. Example: Earthquake magnitude, radon gas value, speed.
- Time Series: A series of regularly recorded observations, such as hourly, daily, weekly or monthly. Example: Weekly ionosphere data.

a) *K-Nearest Neighbors (KNN)* : KNN is a machine learning algorithm used in classification and regression problems.

b) *Support Vector Machine (SVM)* : SVM is a machine learning algorithm used in classification and regression problems that linearly separates two datasets in a way that maximizes the distance between them, and separates them in a three-dimensional space with the help of kernels when the samples cannot be separated linearly.

c) *Artificial Neural Networks (ANN)* : Artificial Neural Networks are algorithms developed by mimicking the learning system of the human brain and are used in regression and classification problems, natural language processing, speech recognition and image recognition problems.

d) *Graph Neural Networks (GNN)* : It is a Neural Network model developed to process data in a graph structure.

e) *Multi-Layer Perceptron* (MLP) : It is a class of "Feed Forward" Artificial Neural Network in which the inputs move unidirectionally towards the output layer.

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f) *Convolutional Neural Networks (CNN):* It is a deep learning architecture often used in image recognition, image processing and analysis.

g) *Recurrent Nuerual Networks (RNN)* : RNN is a deep learning algorithm used in speech recognition, natural language processing and time-dependent problems.

h) *LSTM (Long Short-Term Memory)* : LSTM is a different version of RNNs, often used in time series.

i) *ConvLSTM :* It is a different version of LSTM.

j) *Decision Tree (DT*): Decision Tree is a machine learning algorithm used in classification and regression problems that represents the data with a tree structure as root, nodes and leaves.

k) *Logistic Regression (LR):* Logistic regression is also used to separate variables with two classes (such as as 0 and 1).

l) *Ensemble Learning (Ensemble Methods)* : These are algorithms used in both regression and classification problems that try to get a better result by combining the predictions of more than one model instead of using the prediction result of a single model. There are many methods within this method such as AdaBoost, Random Forests, XGBoost.

m) *Autoregressive Integrated Moving Average (ARIMA)* : It is a method used in time series.

B. Prediction Methods

Scientists examine different natural phenomena in earthquake research. Apart from mathematical methods, the fact that animals have better sense of smell, sight and hearing than humans has led scientists to examine animal behavior [13]. In rare cases, pre-earthquake signals, mathematical calculations, and animal behavior have been shown to be useful methods of predicting earthquakes [14]. Many methods such as mathematical modeling, hydrological methods, ionospheric analyzes, animal behavior to predict earthquakes [15], remote sensing methods using satellite data [16], magnetic field changes, infrared long wave radiation emissions [17] have been studied.

a) Predictions Based on Ionosphere Data

Apart from the relationship of underground resources with earthquakes, the relationship between the effects of the Sun and Moon on Earth and earthquakes is also among the research subjects. Solar events, such as solar winds hitting the earth, have been shown to affect earthquakes around the world [18]. Although the effect of the Moon on earthquakes on the Earth cannot be determined precisely, it has been shown that the Earth's effect is the effect of the Earth in the earthquakes that occur on the Moon, where the Moon is closest to the Earth, and that the planetary positions are related in a few earthquakes in China [19]. There are also studies on the relationship between changes in electrons in the ionosphere and earthquakes. Acoustic waves caused by volcanic eruptions, nuclear explosions, rocket launches, vertical ground changes and shallow earthquakes cause the electron density in the ionosphere to change [20]. In the studies, the increase and decrease of electrons in the ionosphere before and after the earthquake were monitored [21], GPS data [20] and data received from other satellites and the relationships between the electron density in the ionosphere and earthquakes were analyzed [22].

A. Muhammad et. al. used machine learning and deep learning algorithms and architectures ARIMA, LSTM, SVM, ANN to measure the response of radon gas and electrons in the ionosphere to earthquakes and observed that LSTM performed better [23].

M. Akhoondzadeh proposed an Ensemble Methods in which DT, Bagging and RF algorithms are considered together to detect seismo-ionospheric Total Electron Content (TEC) anomalies based on the ionosphere data of the Chile (2010) earthquake [24].

According to E. Senturk, they proved successful in detecting TEC anomalies that may occur due to earthquakes by using a Multi-Neural Network-based hybrid LSTM (N-LSTM) model in the detection of ionospheric anomalies, which they obtained important findings with ionospheric data based on the Nepal (2015) earthquake [25].

A. Akyol et al. used a SVM to examine the relationship of spatio-temporal anomaly with earthquakes using GPS and TEC data [26].

In their study, P. Xiong et al. LR, Ridge Regression, Stochastic Gradient Descent Classifier, Passive Aggressive Classifier, Linear Discriminant Analysis and Quadratic Discriminant Analysis from linear algorithms and Classification and Regression Trees from non-linear algorithms for the identification of pre-earthquake ionosphere disturbances, They used Extra Tree, SVM and NB classifiers, Adaboost, RF, Gradient Boosting Machine, Extreme Gradient Boosting and Light Gradient Boosting Machine algorithms from ensemble learning algorithms, and DNN algorithm from deep learning algorithms. In their study, they observed that the Light Gradient Boosting Machine (LightGBM) model performed better than other models [27].

b) Methods for Changes in Underground Resources

The relationship between the changes in underground resources and earthquakes has an important place. In the studies, the level of groundwater, temperature and geochemical compositions, the stress of the earth's crust were taken into account in investigating the relationships between hydrology and earthquakes, and it was observed that these were associated with earthquakes [28]. In addition, the changes in the count rate of radon gas and the concentration ratios of dissolved elements were observed before the earthquake [29].

T. Haider et. al. applied ANN, MLR and DT algorithms to examine the active-source radon gas anomalies and observed that ANN performed much better in anomaly detection than MLR and DT algorithms [17].

In their study, B. Zmazek et al. used soil radon gas data for earthquake prediction and observed that DT algorithms were better than the regression models they compared [4].

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S. Zhang et al. used the XGBoost algorithm, a decision tree-based algorithm, to analyze abnormal changes in radon gas prior to the Lijiang (Mw 7.0) earthquake. [30].

A. Negarestani et al. used the layered neural network to predict the radon concentration in earthquake prediction [31].

According to A. R. Scorzini, they used LSTM to study hydrological changes in karst aquifers due to a seismic activity. They observed that the use of machine learning approaches for flow prediction in seismic karst areas was successful, although model building was affected by insufficient data [32].

In their study, A. A. Mir et al. used Ensemble Methods such as Boosted Tree (BT), Bagged Cart and Boosted Linear, and SVM with linear and radial kernels and KNN algorithms to predict earthquake probability with anomalies in data affected by seismic activity. As a result of the study, they observed that SVM with radial kernel and BT algorithms gave better results [33].

c) Mathematical Models

In this section, various studies using machine learning and deep learning algorithms are included.

M. A. Murti et al. used SVM, RF, DT and ANN algorithms from machine learning algorithms in their proposed work to distinguish between earthquake vibration and non-earthquake vibrations using acceleration seismic waves, and they observed that ANN gave the best solution to distinguish vibrations [34].

K. M. Asım et al. used ANN, SVM and RF algorithms for the prediction and analysis of short-term low magnitude (3.0, 3.5, 4.0, 4.5 momentum magnitude) seismic activities in the Cyprus region based on the earthquake catalog in their published article, Matthews correlation RF for the magnitudes 3.0 and 3.5 based on the coefficient and their estimations for the magnitudes 4.0 and 4.5 were higher in SVM [35].

In their study, A. Joshi et al. They developed an ensemble learning model with machine learning models such as AdaBoost, XGBoost, DT, LightGBM and Lasso Regression. Using 591 records of 137 earthquakes, they estimated the test earthquake of 6.3 magnitude as 6.0 with an absolute mean prediction error of 0.42. Among these models, the Decision Tree model individually gave the lowest error, but when all models were used for training, they obtained the minimum error [36].

According to M. A. Bilal, they proposed a Batch Normalization Graph Convolutional Neural Network (BNGCNN) model, which includes deep learning models Convolutional Neural Networks and GNN for early earthquake detection. They used multi-station and tri-waveform earthquake data of greater than 3 for Southern California from 2010 to 2015 as a dataset and observed that their method was successful in estimating the magnitude, depth and location of an earthquake compared to the basic model Graph Neural Networks [37].

In their study, H. S. Kuyuk and O. Susumu trained the LSTM network to classify the vibrations within the first one second of the P wave formed in an earthquake with the characteristics of 305 three-component accelerations and one-second P wave recorded in Japan between 2000 and 2018, and the result was They reached an accuracy rate of 98.2% [38].

J. Reyes et al. used ANN to predict earthquakes in Chile, where seismic activity is very high. As a result of the studies, they found that ANN were more successful in the KNN, SVM classifiers they compared [39].

According to P. Xiong et al. they developed a model with Adaboost-based Inverse Boosting Pruning Trees (IBPT) for earthquake prediction with earthquake data above magnitude 6 between 2006 and 2013 and compared this model with eight models including Frequent Pattern Learning (FPL), Generalized Linear Models (GLM), Gradient Boosting Machines (GBM), Deep Neural Networks (DNN), RF, CNN, LR and Naive Bayes (NB). As a result of their study, they observed that IBPT gave the best performance [16].

M. Yousefzadeh et al. compared the performance of DNN, SVM, DT and Shallow Neural Network in predicting the magnitude of an earthquake in the next week. They observed that although DNN and SVM performed well in predicting the high magnitude class, the performance of the DT was better in high and low magnitude predictions [15].

In their study, A. Berhich et al. used LSTM architecture to predict the magnitude, location and time of an upcoming earthquake greater than 6 using data from Japan and the Northern Red Sea. They observed better results than other studies they compared in their paper [40].

M. Mousavi et al. designed a regressor using deep learning algorithms CNN and RNN with raw waveforms recorded at single stations and observed that this method can improve the performance of station-based learning to predict earthquake magnitude [41].

D. S. Parihar et al. designed a seismic classifier with machine learning classifiers to distinguish and classify earthquake signals from seismic noises. They used the classifiers KNN, Maximum Likelihood, ANN and Support SVM and observed that SVM was more performant than other classifiers in their studies [42].

In their study, Y. Essam et al. developed an ANN model using earthquake acceleration, depth, and velocity parameters to predict earthquake ground motions in Terengganu and compared it with the RF algorithm. As a result, they observed that ANN predict earthquake acceleration, depth and velocity better [43].

Z. Bao et al. classified the earthquake magnitude with the CNN model from the data collected from the electromagnetic sensor [44] .

A. Berhich et al. In location-based earthquake predictions covering Morocco, Japan and Turkey, they divided the data set into clusters with the unsupervised learning algorithm of machine learning, K-Means and used deep learning algorithms LSTM, Gated Recurrent Unit (GRU) and these two models as hybrids [45].

A. Varol et al. investigated The Progress of Earthquake Risk Reduction and Recovery in Turkey: From Marmara to Van and they recommended important points on how recovery and mitigation can be performed after an earthquake [46].

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A. Varol proposed research about disaster management, disaster education and disaster awareness. The author compared the education differences between Turkiye and the US in terms of disaster management and education and gave some recommendations [47].

IV. Conclusion

Since the occurrence of earthquakes is random, the exact time of their occurrence is not known today. However, as shown in the studies in this paper, various methods of predicting earthquakes are being studied. In previous years, it was difficult to distinguish earthquake signals from other environmental signals, but as shown in this paper, machine learning and deep learning algorithms can be used to distinguish between mixed signals, estimate earthquake magnitude, and predict earthquakes based on changes in underground resources. In the studies in this paper, many algorithms are used in earthquake prediction methods. In machine learning, instead of accepting the accuracy of only one algorithm against a problem, multiple algorithms should be tested, and their accuracy rates should be compared. The success of deep learning algorithms, which are based on neural networks and are a sub-branch of machine learning, stand out in the majority of studies. Deep learning algorithms are algorithms that can work with large data sets and are capable of self-learning and self-improvement without human intervention. Therefore, it can be expected that deep learning algorithms trained with very large data sets including all factors that may be effective in triggering earthquakes such as underground resources, atmosphere / ionosphere, extraterrestrial factors in the development of early warning systems and earthquake prediction studies in order to prevent the adverse effects of earthquakes will perform well in the prediction of earthquakes.

V. References

[1] K. D. Gunasekera, Rashmin; Ishizawa Escudero, Oscar Anil; Daniell, James Edward; Pomonis, Antonios; Macabuag, Joshua Lee David Clifton; Brand, Johannes; Schaefer, Andreas; Romero, Roberth; Esper, Sarah; Otálora, Samuel González; Khazai, Bijan; Cox, "Global Rapid Post-Disaster Damage Estimation (GRADE) Report February 6 , 2023 Kahramanmaraş Earthquakes Türkiye Report," 2023.

[2] J. Reid, J. Liu, and B. Ananthabhotla, "QuakeCast , an Earthquake Forecasting System Using Ionospheric Anomalies and Machine Learning," 2022.

[3] K. A. Shidiqi, A. Di Paolo, and Á. Choi, "Earthquake exposure and schooling: Impacts and mechanisms," *Econ. Educ. Rev.*, vol. 94, no. June 2022, 2023, doi: 10.1016/j.econedurev.2023.102397.

[4] B. Zmazek, L. Todorovski, S. Džeroski, J. Vaupotič, and I. Kobal, "Application of decision trees to the analysis of soil radon data for earthquake prediction," *Appl. Radiat. Isot.*, vol. 58, no. 6, pp. 697–706, 2003, doi: 10.1016/S0969-8043(03)00094-0.

[5] F. Chollet, *Deep Learning with Python*. Manning Publications, pg.18-23, 2017. ISBN: 9781617294433 [Online]. Available: https://www.manning.com/books/deep-learning-with-python.

[6] G. C. Beroza, M. Segou, and S. Mostafa Mousavi, "Machine learning and earthquake forecasting—next steps," *Nat. Commun.*, vol. 12, no. 1, pp. 10–12, 2021, doi: 10.1038/s41467-021-24952-6.

[7] M. A. Meier *et al.*, "Reliable Real-Time Seismic Signal/Noise Discrimination With Machine Learning," *J. Geophys. Res. Solid Earth*, vol. 124, no. 1, pp. 788–800, 2019, doi: 10.1029/2018JB016661.

[8] B. Gomez and U. Kadri, "Earthquake source characterization by machine learning algorithms applied to acoustic signals," *Sci. Rep.*, vol. 11, no. 1, pp. 1–15, 2021, doi: 10.1038/s41598-021-02483-w.

[9] "International Platform on Earthquake Early Warning Systems (IP-EEWS)." https://en.unesco.org/disaster-risk-reduction/early-warning-systems/IP-EEWS

[10] G. Cremen and C. Galasso, "Earthquake early warning: Recent advances and perspectives," *Earth-Science Rev.*, vol. 205, no. April, p. 103184, 2020, doi: 10.1016/j.earscirev.2020.103184.

[11] R. M. Allen and D. Melgar, "Earthquake early warning: Advances, scientific challenges, and societal needs," *Annu. Rev. Earth Planet. Sci.*, vol. 47, pp. 361–388, 2019, doi: 10.1146/annurev-earth-053018-060457.

[12] A. Galkina and N. Grafeeva, "Machine learning methods for earthquake prediction: A survey," *CEUR Workshop Proc.*, vol. 2372, no. April, pp. 25–32, 2019.

[13] F. ti Fan, "Can animals predict earthquakes?: Bio-sentinels as seismic sensors in communist China and beyond," *Stud. Hist. Philos. Sci. Part A*, vol. 70, pp. 58–69, 2018, doi: 10.1016/j.shpsa.2018.05.009.

[14] M. Bhatia, T. A. Ahanger, and A. Manocha, "Artificial intelligence based real-time earthquake prediction," *Eng. Appl. Artif. Intell.*, vol. 120, no. December 2022, p. 105856, 2023, doi: 10.1016/j.engappai.2023.105856.

[15] M. Yousefzadeh, S. A. Hosseini, and M. Farnaghi, "Spatiotemporally explicit earthquake prediction using deep neural network," *Soil Dyn. Earthq. Eng.*, vol. 144, no. February, p. 106663, 2021, doi: 10.1016/j.soildyn.2021.106663.

[16] P. Xiong *et al.*, "Towards advancing the earthquake forecasting by machine learning of satellite data," *Sci. Total Environ.*, vol. 771, p. 145256, 2021, doi: 10.1016/j.scitotenv.2021.145256.

[17] T. Haider *et al.*, "Identification of radon anomalies induced by earthquake activity using intelligent systems," *J. Geochemical Explor.*, vol. 222, no. December 2020, p. 106709, 2021, doi: 10.1016/j.gexplo.2020.106709.

[18] M. Tavares and A. Azevedo, "Influences of Solar Cycles on Earthquakes," *Nat. Sci.*, vol. 03, no. 06, pp. 436–443, 2011, doi: 10.4236/ns.2011.36060.

[19] V. G. Kolvankar and B. Atomic, "Lunar periodicities and earthquakes," vol. 56, no. September 2010, 2010.

[20] E. Calais and J. B. Minster, "GPS, earthquakes, the ionosphere, and the Space Shuttle," *Phys. Earth Planet. Inter.*, vol. 105, no. 3–4, pp. 167–181, 1998, doi: 10.1016/S0031-9201(97)00089-7.

[21] J. Ahmed, M. Shah, W. A. Zafar, M. A. Amin, and T. Iqbal, "Seismoionospheric anomalies associated with earthquakes from the analysis of the ionosonde data," *J. Atmos. Solar-Terrestrial Phys.*, vol. 179, no. March, pp. 450–458, 2018, doi: 10.1016/j.jastp.2018.10.004.

[22] Y. He, X. Zhao, D. Yang, Y. Wu, and Q. Li, "A study to investigate the relationship between ionospheric disturbance and seismic activity based on Swarm satellite data," *Phys. Earth Planet. Inter.*, vol. 323, 2022, doi: 10.1016/j.pepi.2021.106826.

[23] A. Muhammad, F. Külahcı, and S. Birel, "Investigating radon and TEC anomalies relative to earthquakes via AI models," *J. Atmos. Solar-Terrestrial Phys.*, vol. 245, no. September 2022, p. 106037, 2023, doi: 10.1016/j.jastp.2023.106037.

[24] M. Akhoondzadeh, "Decision Tree, Bagging and Random Forest methods detect TEC seismo-ionospheric anomalies around the time of the Chile, (M w = 8.8) earthquake of 27 February 2010," *Adv. Sp. Res.*, vol. 57, no. 12, pp. 2464–2469, 2016, doi: 10.1016/j.asr.2016.03.035.

[25] E. Şentürk, M. Saqib, and M. A. Adil, "A Multi-Network based Hybrid LSTM model for ionospheric anomaly detection: A case study of the Mw 7.8 Nepal earthquake," *Adv. Sp. Res.*, vol. 70, no. 2, pp. 440–455, 2022, doi: 10.1016/j.asr.2022.04.057.

[26] A. A. Akyol, O. Arikan, and F. Arikan, "A Machine Learning-Based Detection of Earthquake Precursors Using Ionospheric Data," *Radio Sci.*, vol. 55, no. 11, 2020, doi: 10.1029/2019RS006931.

[27] P. Xiong, C. Long, H. Zhou, R. Battiston, X. Zhang, and X. Shen, "Identification of electromagnetic pre-earthquake perturbations from the demeter data by machine learning," *Remote Sens.*, vol. 12, no. 21, pp. 1–27, 2020, doi: 10.3390/rs12213643.

[28] X. Yan, Z. Shi, G. Wang, H. Zhang, and E. Bi, "Detection of possible hydrological precursor anomalies using long short-term memory: A case study of the 1996 Lijiang earthquake," *J. Hydrol.*, vol. 599, no. January, p. 126369, 2021, doi:

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10.1016/j.jhydrol.2021.126369.

[29] A. Skelton *et al.*, "Changes in groundwater chemistry before two consecutive earthquakes in Iceland," *Nat. Geosci.*, vol. 7, no. 10, pp. 752–756, 2014, doi: 10.1038/NGEO2250.

[30] S. Zhang, Z. Shi, G. Wang, R. Yan, and Z. Zhang, "Application of the extreme gradient boosting method to quantitatively analyze the mechanism of radon anomalous change in Banglazhang hot spring before the Lijiang Mw 7.0 earthquake," *J. Hydrol.*, vol. 612, no. PC, p. 128249, 2022, doi: 10.1016/j.jhydrol.2022.128249.

[31] A. Negarestani, S. Setayeshi, M. Ghannadi-Maragheh, and B. Akashe, "Layered neural networks based analysis of radon concentration and environmental parameters in earthquake prediction," *J. Environ. Radioact.*, vol. 62, no. 3, pp. 225–233, 2002, doi: 10.1016/S0265-931X(01)00165-5.

[32] A. R. Scorzini, M. Di Bacco, G. De Luca, and M. Tallini, "Deep learning for earthquake hydrology? Insights from the karst Gran Sasso aquifer in central Italy," *J. Hydrol.*, vol. 617, no. PB, p. 129002, 2023, doi: 10.1016/j.jhydrol.2022.129002.

[33] A. A. Mir *et al.*, "Anomalies Prediction in Radon Time Series for Earthquake Likelihood Using Machine Learning-Based Ensemble Model," *IEEE Access*, vol. 10, pp. 37984–37999, 2022, doi: 10.1109/ACCESS.2022.3163291.

[34] M. A. Murti, R. Junior, A. N. Ahmed, and A. Elshafie, "Earthquake multi-classification detection based velocity and displacement data filtering using machine learning algorithms," *Sci. Rep.*, vol. 12, no. 1, pp. 1–12, 2022, doi: 10.1038/s41598-022-25098-1.

[35] K. M. Asim, S. S. Moustafa, I. A. Niaz, E. A. Elawadi, T. Iqbal, and F. Martínez-Álvarez, "Seismicity analysis and machine learning models for short-term low magnitude seismic activity predictions in Cyprus," *Soil Dyn. Earthq. Eng.*, vol. 130, no. April 2019, 2020, doi: 10.1016/j.soildyn.2019.105932.

[36] A. Joshi, C. Vishnu, and C. K. Mohan, "Early detection of earthquake magnitude based on stacked ensemble model," *J. Asian Earth Sci. X*, vol. 8, no. August, p. 100122, 2022, doi: 10.1016/j.jaesx.2022.100122.

[37] M. A. Bilal, Y. Ji, Y. Wang, M. P. Akhter, and M. Yaqub, "Early Earthquake Detection Using Batch Normalization Graph Convolutional Neural Network (BNGCNN)," *Appl. Sci.*, vol. 12, no. 15, 2022, doi: 10.3390/app12157548.

[38] H. S. Kuyuk and O. Susumu, "Real-time classification of earthquake using deep learning," *Procedia Comput. Sci.*, vol. 140, pp. 298–305, 2018, doi: 10.1016/j.procs.2018.10.316.

[39] J. Reyes, A. Morales-Esteban, and F. Martínez-Álvarez, "Neural networks to predict earthquakes in Chile," *Appl. Soft Comput. J.*, vol. 13, no. 2, pp. 1314–1328, 2013, doi: 10.1016/j.asoc.2012.10.014.

[40] A. Berhich, F. Z. Belouadha, and M. I. Kabbaj, "An attention-based LSTM network for large earthquake prediction," *Soil Dyn. Earthq. Eng.*, vol. 165, no. June 2022, p. 107663, 2023, doi: 10.1016/j.soildyn.2022.107663.

[41] S. M. Mousavi and G. C. Beroza, "A Machine-Learning Approach for Earthquake Magnitude Estimation," *Geophys. Res. Lett.*, vol. 47, no. 1, pp. 1–7, 2020, doi: 10.1029/2019GL085976.

[42] D. S. Parihar, R. Ghosh, A. Akula, S. Kumar, and H. K. Sardana, "Machine Learning Based Comparative Analysis for the Classification of Earthquake Signals BT - Proceedings of the International Conference on Computing and Communication Systems," 2018, pp. 369–376.

[43] Y. Essam, P. Kumar, A. N. Ahmed, M. A. Murti, and A. El-Shafie, "Exploring the reliability of different artificial intelligence techniques in predicting earthquake for Malaysia," *Soil Dyn. Earthq. Eng.*, vol. 147, no. October 2020, p. 106826, 2021, doi: 10.1016/j.soildyn.2021.106826.

[44] Z. Bao, J. Zhao, P. Huang, S. Yong, and X. Wang, "A deep learning-based electromagnetic signal for earthquake magnitude prediction," *Sensors*, vol. 21, no. 13, 2021, doi: 10.3390/s21134434.

[45] A. Berhich, F. Z. Belouadha, and M. I. Kabbaj, "A location-dependent earthquake prediction using recurrent neural network algorithms," *Soil Dyn. Earthq. Eng.*, vol. 161, 2022, doi: 10.1016/j.soildyn.2022.107389.

[46] A. Varol, J. Enia, "The Progress of Earthquake Risk Reduction and Recovery in Turkey: From Marmara to Van, 6th International Disaster and Risk Conference", IDRC Davos 2016, 28 August – 1 September 2016, Davos, Switzerland.

[47] A. Varol, "Disaster Management, Disaster Education and Disaster Awareness: The Case of America, Fırat University Journal of Social Sciences", 2019, vol. 29, no 1, p. 193-204 doi: 10.18069/firatsbed.538678

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